

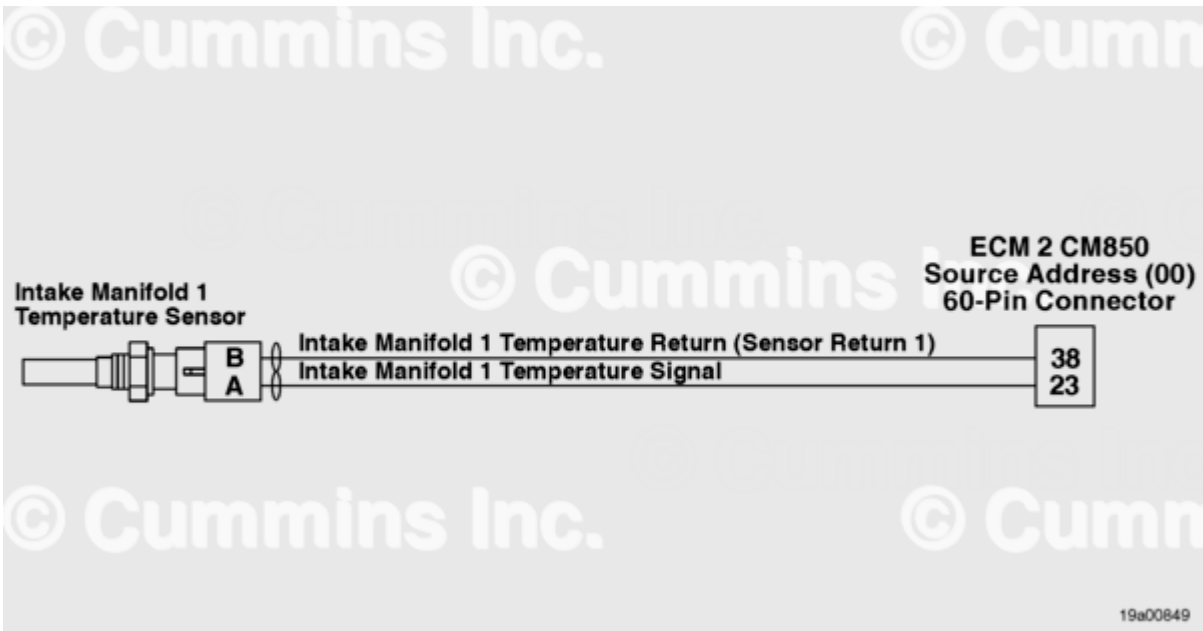
Fault Code: 153

Intake Manifold 1 Temperature Sensor Circuit - Voltage Above Normal or Shorted to High Source

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Overview

Codes	Reason	Effect
Fault Code: 153 PID(P): P105 SPN: 105 FMI: 3/3 Lamp: Amber SRT:	High signal voltage detected at intake manifold 1 temperature sensor circuit.	None on performance.



Intake Manifold 1 Temperature Sensor Circuit

Circuit Description

The intake manifold temperature sensor is a variable resistor sensor used by the ECM to monitor the intake manifold temperature. The intake manifold temperature sensor has two circuits: signal, and return circuits. The signal voltage indicates the intake manifold temperature.

Component Location

The intake manifold 1 temperature sensor is located in the left bank rear air intake manifold.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the generator set controller is active or when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the intake manifold 1 temperature sensor signal voltage is out of range high.

Action Taken When The Fault Code Is Active

- The generator set controller displays a warning fault immediately when the diagnostics runs and fails.
- A default value is used for the intake manifold temperature reading.
- Engine torque will be reduced if the engine is operated for an extended period of time with this fault active.

Conditions For Clearing The Fault Code

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the warning indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

There are multiple ECMs for the engine models included in this manual. Each ECM has an individual source address that displays when Cummins® electronic service tool or equivalent is connected. When troubleshooting a fault code, use the source address displayed in Cummins® electronic service tool or equivalent to determine which ECM and circuit is affected.

Possible causes of this fault code include:

- Malfunctioning Intake Manifold Temperature Sensor
- Malfunctioning or damaged engine wiring harness.

